

Easy-Moderate

Mathematics Advanced12

Practice Questions
(Paper 1)

Name:_____

1. Rationalise the denominator : $\frac{2+\sqrt{3}}{8-2\sqrt{3}}$ (Show your complete solutions)

$$\begin{aligned} & \frac{(2+\sqrt{3})}{(8-2\sqrt{3})} \times \frac{(8+2\sqrt{3})}{(8+2\sqrt{3})} \\ &= \frac{16+4\sqrt{3}+8\sqrt{3}+6}{64-12} \\ &= \frac{22+12\sqrt{3}}{52} \\ &= \frac{11+6\sqrt{3}}{26} \end{aligned}$$

2. Differentiate the following with respect to x .

(i) $\frac{1}{3x}$

$$\begin{aligned} \frac{d}{dx}(3x)^{-1} &= -(3x)^{-2}(3) \\ &= -\frac{3}{9x^2} \\ &= -\frac{1}{3x^2} \end{aligned}$$

(ii) $e^x \cos x$

$$\begin{aligned} \frac{dy}{dx} &= e^x(-\sin x) + \cos x(e^x) \\ &= e^x(\cos x - \sin x) \end{aligned}$$

(iii) $\ln(\sin x)$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos x}{\sin x} \\ &= \cot x \end{aligned}$$

(iv) $(3x^2 - 5x)^6$

$$\begin{aligned} \frac{dy}{dx} &= 6(3x^2 - 5x)^5(6x - 5) \\ &= 6x(6x - 5)(3x - 5)^5 \end{aligned}$$

3. Find the limiting sum of $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$

$$a = 1, \quad r = -\frac{1}{3}$$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1}{1 - -\frac{1}{3}} \\ &= \frac{3}{4} \end{aligned}$$

4. A function is defined by $f(x) = x^3 - 3x^2 - 9x + 22$.
 (i) Find the stationary points and their nature.

$$f(x) = x^3 - 3x^2 - 9x + 22$$

$$f'(x) = 3x^2 - 6x - 9$$

Let $f'(x) = 0$ to find stationary points

$$3(x^2 - 2x - 3) = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\therefore x = 3 \text{ and } x = -1$$

$$\begin{aligned} \text{When } x = 3, y &= 3^3 - 3(3)^2 - 9 \times 3 + 22 \\ &= -5 \end{aligned}$$

$$\begin{aligned} \text{When } x = -1, y &= (-1)^3 - 3(-1)^2 - 9 \times (-1) + 22 \\ &= 27 \end{aligned}$$

$\therefore$ Stationary points: (3, -5) and (-1, 27)

Using 2nd derivative (or table) to find nature of the stationary points.

$$f''(x) = 6x - 6$$

$$\text{At } (3, -5)$$

$$\begin{aligned} f''(3) &= 6(3) - 6 \\ &= 12 \end{aligned}$$

> 0 (concave up)

$\therefore$ (Local) Minimum turning point at (3, -5)

$$\text{At } (-1, 27)$$

$$\begin{aligned} f''(-1) &= 6(-1) - 6 \\ &= -12 \end{aligned}$$

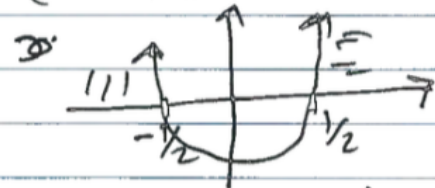
< 0 (concave down)

$\therefore$ (Local) Maximum turning point at (-1, 27)

5. Find the domain of the function $f(x) = \frac{1}{\sqrt{4x^2 - 1}}$.

$$4x^2 - 1 > 0$$

$$(2x+1)(2x-1) > 0$$



$$x < -\frac{1}{2} \quad x > \frac{1}{2}$$

6. Simplify $\frac{x^2y - xy^2}{x^2 - y^2}$.

$$\frac{x^2y - xy^2}{x^2 - y^2} = \frac{xy(x-y)}{(x-y)(x+y)}$$

$$= \frac{xy}{x+y}$$

7. Find the equation of the tangent to the curve $f(x) = e^{1-2x}$ at the point where $x = \frac{1}{2}$. Write the equation in general form.

$$y = e^{1-2x}$$

$$\frac{dy}{dx} = e^{1-2x} (-2)$$

$$\text{gradient of tangent } m_{x=\frac{1}{2}} = e^{1-2 \times \frac{1}{2}} (-2)$$

$$= -2$$

$$x = \frac{1}{2} \quad y = 1$$

so using gradient point of tangent to the curve

$$y - 1 = -2(x - \frac{1}{2})$$

$$y - 1 = -2x + 1$$

$$2x + y - 2 = 0$$

8. Find the equation of the tangent to the curve $y = x^3 - 4x^2$ at the point $(2, -8)$.

$$y - y_1 = m(x - x_1)$$

$$y' = 3x^2 - 8x$$

$$\text{at } x = 2$$

$$m = 3(2)^2 - 8(2)$$

$$= -4$$

$$y - (-8) = -4(x - 2)$$

$$y + 8 = -4x + 8$$

$$y = -4x$$

9. Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$.

$$\frac{1}{2} [\tan 2x]_0^{\frac{\pi}{8}} \quad \checkmark \quad \text{lim}$$

$$= \frac{1}{2} [\tan \frac{\pi}{4} - \tan 0]$$

$$= \frac{1}{2} [1 - 0]$$

$$= \frac{1}{2} \quad \checkmark \quad \text{lim}$$

10. Find $\int 3 + \sin 4\pi x \, dx$

$$\int 3 + \sin 4\pi x \, dx = 3x - \frac{\cos 4\pi x}{4\pi} + C$$

11. Solve $\ln(7x - 12) = 2 \ln x$.

$$\begin{array}{l|l} \ln(7x-12) = \ln(x^2) & 7(3)-12=9 \\ 7x-12 = x^2 & >0 \therefore \text{True} \\ x^2 - 7x + 12 = 0 & 7(4)-12=16 \\ (x-3)(x-4)=0 & >0 \therefore \text{True} \\ \therefore x=3, x=4 & \end{array}$$

12. Find the equation of the tangent to the curve $y = e^{x+2}$ at the point $A(-2, 1)$.

$$\begin{array}{l} y' = e^{x+2} \\ \text{when } x = -2 \\ y' = e^{-2+2} \\ = e^0 \\ y' = 1 \quad \checkmark \end{array} \quad \begin{array}{l} y-1 = 1(x-(-2)) \\ y-1 = x+2 \\ y = x+3 \quad \checkmark \end{array}$$

13. Differentiate $y = \frac{x^2}{1-e^x}$

$$y' = \frac{v u' - u v'}{v^2}$$

$$u = x^2, u' = 2x$$

$$v = 1 - e^x, v' = -e^x$$

$$y' = \frac{2x(1 - e^x) + x^2 e^x}{(1 - e^x)^2}$$

14. Express $\frac{3+\sqrt{2}}{6+\sqrt{2}}$ with a rational denominator.

$$\begin{aligned} & \text{Sol) } \frac{3+\sqrt{2}}{6+\sqrt{2}} \times \frac{6-\sqrt{2}}{6-\sqrt{2}} \\ &= \frac{18 - 3\sqrt{2} + 6\sqrt{2} - 2}{36 - 2} \\ &= \frac{16 + 3\sqrt{2}}{34} \end{aligned}$$

15. Solve $2 \sin \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

$$\begin{aligned} 2 \sin \theta - 1 &= 0 \\ \sin \theta &= \frac{1}{2} \\ \therefore \theta &= \frac{\pi}{6}, \frac{5\pi}{6} \end{aligned}$$

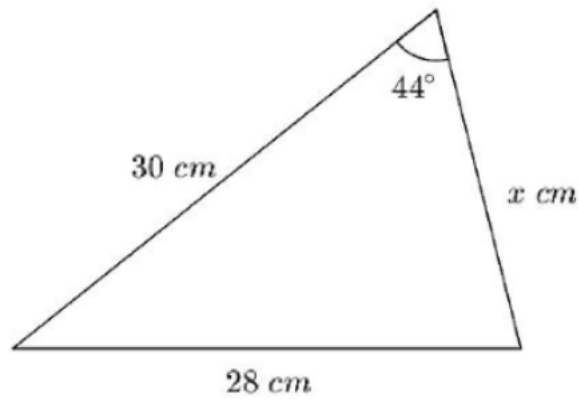
16. Find $\int \frac{x}{x^2-1} dx$

$$\begin{aligned} & \int \frac{u}{u^2-1} du \\ &= \frac{1}{2} \int \frac{2u}{u^2-1} du \\ &= \frac{1}{2} \ln(u^2-1) + C \end{aligned}$$

17. Find $\int \frac{5x^2}{x^3+1} dx$

$$\begin{aligned} &= \frac{5}{3} \int \frac{3x^2}{x^3+1} dx \\ &= \frac{5}{3} \ln|x^3+1| + C \end{aligned}$$

18. The area of the triangle shown is 250 cm^2 .



What is the value of x to the nearest whole number?

$$A = \frac{1}{2} ab \sin C$$

$$250 = \frac{1}{2} (x)(30) \sin 44 \quad (1)$$

$$x = 250 \div 15 \sin 44 \quad (1)$$

$$x = 23.99 \text{ cm}$$

$$x = 24 \text{ cm} \quad (1)$$

19. Find $\frac{d}{dx}(x^3 \tan 2x)$

$$u = x^3 \quad v = \tan 2x$$

$$u' = 3x^2 \quad v' = 2 \sec^2 2x$$

$$\left. \begin{aligned} \frac{d}{dx}(x^3 \tan 2x) &= 3x^2 \tan 2x + 2x^3 \sec^2 2x \\ &= x^2 (3 \tan 2x + 2x \sec^2 2x) \end{aligned} \right\}$$

20. Find $\int \frac{2x+2}{4x^2+8x+1} dx$.

$$\int \frac{2x+2}{4x^2+8x+1} dx$$

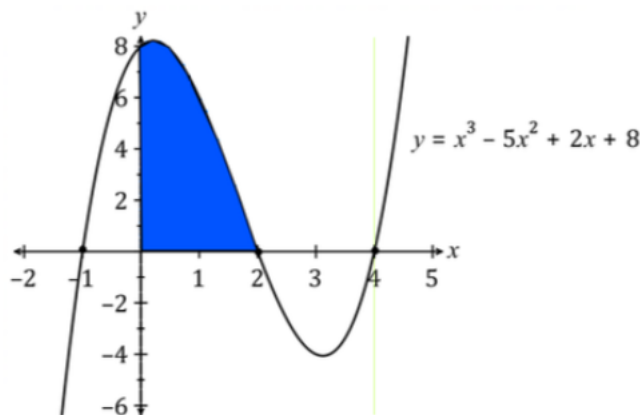
$$f(x) = 4x^2 + 8x + 1$$

$$f'(x) = 8x + 8$$

$$= \frac{1}{4} \int \frac{4(2x+2)}{4x^2+8x+1} dx$$

$$= \frac{1}{4} \ln |4x^2+8x+1| + C$$

21.



Find the shaded area enclosed by the curve $y = x^3 - 5x^2 + 2x + 8$ and the coordinate axes.

$$A = \int_0^2 x^3 - 5x^2 + 2x + 8 dx$$

$$= \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + x^2 + 8x \right]_0^2$$

$$= \left(\frac{1}{4}(2)^4 - \frac{5}{3}(2)^3 + 2^2 + 8 \times 2 \right) - 0$$

$$= \frac{32}{3} = 10 \frac{2}{3}$$

22. Find $\int 5x^2(x^3+1)^{10} dx$.

$$\frac{d}{dx} (x^3+1)^{11} = 11(x^3+1)^{10} \times 3x^2$$

$$= 33x^2(x^3+1)^{10}$$

$$\therefore \int 5x^2(x^3+1)^{10} dx = \frac{5}{33} (x^3+1)^{11} + C$$